

Endogenous Trade Fragmentation

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Abstract

Each firm that leaves a cross-bloc supply relationship reduces the trade surplus that deters disruption. The exit raises the hazard borne by firms that remain. I place this private sourcing externality in a multisector model with route thickness and endogenous friendly capacity. The local response equals $(I - J)^{-1}\mathbf{b}$, where J collects sector best responses. Nondegenerate folds create sectoral jumps that aggregate trade shares can conceal. In the reference calibration, $\varrho(J) = 0.196$ and equilibrium feedback amplifies the critical-sector response by 1.24. Perfect-foresight and adaptive dynamics select the same terminal branches, although anticipation accelerates adjustment during a temporary confrontation. Inventory and emergency substitution lower the critical disruption loss from 150 percent to 75 or 33.3 percent in the reported buffer cases. A maintenance subsidy raises welfare when the value of independent sourcing lies below the deterrence externality.

JEL codes: F13, F51, F52, D85, C62.

Keywords: fragmentation, friend-shoring, supply chains, strategic complementarity, tipping points, deterrence.

1 Introduction

China supplied about 22 percent of United States goods imports in 2017 and about 14 percent in 2023. Total imports stayed high over those years. The gap moved to Vietnam, Mexico, and other countries positioned between the blocs (Alfaro and Chor, 2023; Freund et al., 2023). The movement was uneven: trade between geopolitical blocs has slowed relative to trade within them since Russia invaded Ukraine in February 2022, with the sharpest breaks in strategically sensitive products (Gopinath et al., 2025). The reallocation is a rewiring of the network along political lines, concentrated exactly where substitution is hardest.

The question is whether that rewiring stops on its own or feeds on itself. A one-time increase in geopolitical risk may produce a smooth repricing. It may instead begin an unraveling in which one firm's exit makes the next exit more attractive. Governments support friend-shoring because they believe private firms undervalue resilience. Quantitative studies price fragmentation counterfactuals conditional on an imposed split and find costs ranging from 0.2 to 7 percent of global GDP (Aiyar et al., 2023; Baqaee et al., 2024; Attinasi et al., 2023). Endogenous sourcing changes the forecast and can reverse the sign of policy.

The model is deliberately spare. In each sector, firms choose between a cheap cross-bloc supplier exposed to disruption and a politically aligned supplier charging an idiosyncratic premium. Escalation destroys surplus on the trade that remains. A government weighs that loss against the political benefit of disruption, so each private exit can weaken deterrence for other firms. Route thinning raises disruption losses as trade leaves. Expansion of aligned capacity lowers future switching premia. The model keeps these mechanisms separate because they imply different empirical moments and policy wedges.

Strategic complementarity is a vector object. At a locally stable equilibrium, the response to a hazard shifter is $(I - J)^{-1}\mathbf{b}$. Within-sector comparisons recover direct threshold responses. State variation and cross-sector spillover moments recover J , since sectors differ in premium distributions and aggregate loadings.

Nonlinear amplification can produce a fold in a critical sector. The theorem combines tangency, transversality, and nonzero curvature. A maximum best-response slope above one supplies one part of this system. The reference composite map yields a band from 11.4 to 13.1 percent. Numerical continuation finds every root on four grids and solves the fold conditions to residual 2.2×10^{-11} . A temporary hazard increase creates permanent fragmentation when it moves the state across the unstable branch.

The policy wedge compares the hazard and route externalities imposed on firms that stay with the capacity benefit delivered to firms that switch. A government value v_m for independent sourcing subtracts from this wedge. The reference maintenance wedge is 6.4 percent before v_m and instrument costs. The threshold $v_m < \tau_m^*$ determines the sign of corrective policy.

The resilience literature studies sourcing against disruption risk (Grossman et al., 2023). Network models generate discontinuous fragility through endogenous links (Elliott et al., 2022). Trade-and-conflict evidence establishes that commerce can deter escalation (Martin et al., 2008; Jackson and Nei, 2015). Mayer et al. (2025) place diplomacy and escalation inside a quantitative trade model. Governments choose trade policy in their model, and decoupling raises escalation by shrinking its opportunity cost. Here private firms choose suppliers. Each firm ignores the deterrence value of its own trade link. That private margin produces a multisector Jacobian, sectoral folds that aggregate data can hide, and a welfare wedge whose sign changes across branches. Route thickness and friendly capacity determine how the wedge changes as firms leave.

Recent measurement documents reallocation along political lines and its concentration in sensitive sectors (Gopinath et al., 2025; Fernández-Villaverde et al., 2024; Alfaro and Chor, 2023). Section 6 combines those facts with state variation and cross-sector spillover moments to estimate the feedback parameters.

The quantitative exercise reports phase diagrams and their numerical residuals. Welfare accounting includes switching premia and expected disruption losses. The independence value enters as a policy primitive. Public-data moments discipline every parameter that controls the Jacobian or the fold conditions.

2 The Model

The analysis separates general sign results from affine-hazard calculations. Strategic complementarity follows from the escalation-benefit density and the slope of destroyed trade surplus. The quantitative section uses a uniform benefit distribution to obtain an affine hazard. Numerical continuation then establishes fold existence and global branch structure.

2.1 Firms and the sourcing choice

There are two blocs and M sectors. In sector m , a unit mass of Home firms currently source a key intermediate input from a cross-bloc supplier, and sector m carries a share w_m of cross-bloc intermediate trade. Time is discrete and annual. Each firm chooses a sourcing mode: *cross-bloc*, keeping the incumbent supplier, or *friend-shored*, replacing it with an in-bloc supplier of the same input.

Friend-shoring costs firm i a flow premium g_i , expressed ad valorem on the input: the in-bloc supplier is dearer or must first be built. The premium is idiosyncratic, $g_i \sim H_m$, lognormal with median $\bar{g}_m$ and log dispersion s . Heterogeneity stands in for relationship capital and product-specific reshoring costs. A fraction f_m^0 of firms switched before geopolitics entered the calculus, for wages or diversification, and I take that margin as given.

Cross-bloc sourcing is cheap but exposed. With hazard π_m per year an export control or embargo severs the link. The firm then bears the flow-equivalent disruption loss λ_m and moves to an aligned supplier. Suppose a CES input bundle combines the cross-bloc variety with a composite substitute. Let σ_m denote the substitution elasticity and let s_m denote the initial cost share. Removing the variety raises the input price index by $(1 - s_m)^{1/(1-\sigma_m)}$. A disruption lasting D_m years costs

$$\lambda_m^0 = \left[(1 - s_m)^{\frac{1}{1-\sigma_m}} - 1 \right] D_m, \quad (1)$$

which rises when short-run substitution becomes harder or recovery takes longer. Inventory and emergency suppliers limit the zero-quantity calculation. If inventory covers I_m years and emergency sourcing replaces fraction e_m of the exposed input, the loss becomes

$$\lambda_m^0(I_m, e_m) = \left[(1 - s_m(1 - e_m))^{\frac{1}{1-\sigma_m}} - 1 \right] [D_m - I_m]_+. \quad (2)$$

Short-run production disruptions provide the relevant discipline. Barrot and Sauvagnat (2016), Boehm et al. (2019), and Carvalho et al. (2021) measure persistent output effects after input-specific shocks and recovery horizons near one year. Input-output shares come from OECD ICIO (OECD, 2026). Long-run trade elasticities provide an upper sensitivity bound rather than the preferred substitution input.

The sourcing choice reduces exactly to a threshold rule.

Lemma 1 (The flow comparison is exact). *Suppose a disrupted firm switches perforce and firms discount at any $\beta_f \in (0, 1)$. In a stationary environment, staying is optimal if and only if the*

flow comparison holds:

$$\pi_m \lambda_m \leq g_i^{\text{eff}},$$

where g_i^{eff} is the effective premium defined below. Forward-looking and myopic thresholds coincide.

The logic is that switching is an absorbing state whose cost is the premium forever, while staying pays the expected disruption loss and, upon disruption, inherits the same absorbing state, so the continuation values cancel and the flow comparison survives intact (Appendix A). Along transition paths a small option-value correction appears. Appendix C bounds it and connects the selection issue to Krugman (1991) and Matsuyama (1991).

2.2 The feedback, derived

Let $F_m \in [0, 1]$ denote the fragmented share of sector m 's cross-bloc trade, and let $F = \sum_m w_m F_m$.

The hazard. Disputes arrive with probability δ_m . Conditional on a dispute, escalation yields benefit b with distribution G_m and density g_m^b . It destroys surplus

$$\mathcal{C}_m(F_m, F) = \vartheta_m \bar{S}_m(1 - F_m) + \vartheta_a \bar{S}(1 - F). \quad (3)$$

The government escalates when $b \geq \mathcal{C}_m$. An additional systemic component $\eta_a F$ captures disruptions originating outside the sector-specific dispute. The hazard is

$$\pi_m(F_m, F) = \delta_m [1 - G_m(\mathcal{C}_m(F_m, F))] + \eta_a F. \quad (4)$$

Under a uniform benefit distribution and an interior escalation probability, the quantitative specification sets $\vartheta_a = 0$ in $\mathcal{C}_m$ and retains cross-sector exposure through $\eta_a F$. Normalization of the intercept gives

$$\pi_m(F_m, F) = \delta_m [1 - \theta(1 - F_m)] + \eta_a F, \quad (5)$$

where θ is the sector deterrence share. The quantitative exercise uses the affine equation. The sign results use (4). In particular,

$$\frac{\partial \pi_m}{\partial F_n} = -\delta_m g_m^b(\mathcal{C}_m) \frac{\partial \mathcal{C}_m}{\partial F_n} + \eta_a w_n > 0 \quad (6)$$

whenever remaining trade raises the cost of escalation. Uniformity is required for linearity, not for strategic complementarity.

The loss: routes thin as trade leaves. Shared infrastructure spreads disruption: alternative routings, insurance pools, spot markets, and buffer inventories are thick when many firms use the corridor and thin when few do. I let the loss rise linearly with the sector's fragmentation,

$$\lambda_m(F_m) = \lambda_m^0 (1 + \xi F_m), \quad (7)$$

with $\xi \geq 0$ the thinning elasticity.

The premium: friendly capacity scales. The in-bloc supply base gets cheaper as it gets bigger: entry, learning, and fixed-cost spreading in the destination countries compress the premium. The effective premium is

$$g_i^{\text{eff}} = g_i (1 - \chi F_m), \quad (8)$$

with $\chi \geq 0$ the capacity-scale elasticity. This is the channel the post-2018 reallocation to Vietnam and Mexico makes visible (Alfaro and Chor, 2023): each cohort of switchers builds the capacity that lowers the next cohort's cost.

Combining (5)–(8) with Lemma 1, firm i friend-shores when g_i falls below the threshold

$$\hat{g}_m(F_m, F) = \frac{\pi_m(F_m, F) \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}, \quad (9)$$

and the sector's fragmented share consistent with individual choices is

$$F_m = T_m(F_m, F) \equiv f_m^0 + (1 - f_m^0) H_m(\hat{g}_m(F_m, F)). \quad (10)$$

Proposition 1 (Strategic complementarity). *Suppose $g_m^b(C_m) > 0$, $h_m(\hat{g}_m) > 0$, $\partial C_m / \partial F_n < 0$, and $1 - \chi F_m > 0$. Then T_m is increasing in every F_n . Under the affine hazard, the own-sector threshold derivative is*

$$\frac{\partial \hat{g}_m}{\partial F_m} = \underbrace{\frac{\delta_m \theta \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}}_{\text{deterrence erosion}} + \underbrace{\frac{\pi_m \lambda_m^0 \xi}{1 - \chi F_m}}_{\text{route thinning}} + \underbrace{\frac{\pi_m \lambda_m^0 (1 + \xi F_m) \chi}{(1 - \chi F_m)^2}}_{\text{capacity scale}} > 0,$$

The cross-sector derivative is

$$\frac{\partial \hat{g}_m}{\partial F_n} = \frac{\eta_a w_n \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m} > 0$$

when $\eta_a > 0$. If $\theta = \xi = \chi = \eta_a = 0$, the threshold is independent of fragmentation and the equilibrium is unique.

Deterrence erosion is a security externality. Route thinning and capacity scale are market externalities of the kind studied by Cooper and John (1988). Their relative size determines whether a fold exists and which policy margin is corrective.

The static objects are now in place: a threshold economy in each sector, coupled through the hazard and the aggregate. What remains is to characterize the fixed points of (10), and that is where the multiplier and the tipping structure live.

3 Equilibrium: Multipliers, Tipping, Hysteresis

3.1 Structure and the fragmentation multiplier

An equilibrium is a vector $\mathbf{F} = (F_1, \dots, F_M)$ satisfying $\mathbf{F} = \mathbf{T}(\mathbf{F}, z)$. Let $J = D_{\mathbf{F}}\mathbf{T}$ denote the Jacobian, with element $J_{mn} = \partial T_m / \partial F_n$.

Proposition 2 (Multisector equilibrium response). *The continuous map $\mathbf{T}$ has an equilibrium on $\prod_m [f_m^0, 1]$. At an equilibrium with spectral radius $\varrho(J) < 1$, the local response to a scalar primitive z is*

$$\frac{d\mathbf{F}^*}{dz} = (I - J)^{-1} \mathbf{b}_z, \quad \mathbf{b}_z \equiv D_z \mathbf{T}. \quad (11)$$

All elements of $(I - J)^{-1}$ are nonnegative. If the system is sectorally decoupled, (11) reduces to

$$\frac{dF_m^*}{dz} = \frac{b_{mz}}{1 - J_{mm}}.$$

The scalar formula is exact only in that special case.

Corollary 1 (Direct effects and network effects). *Within-sector comparisons that hold $\mathbf{F}$ fixed identify components of $\mathbf{b}_z$. Joint state variation and cross-sector spillover moments identify J , since sectors differ in H_m , λ_m^0 , and their loadings on aggregate fragmentation. These moments then determine $(I - J)^{-1}$.*

At the reference scenario, $\varrho(J) = 0.196$. The critical-sector direct derivative with respect to δ is 0.885 and its total derivative is 1.101. The corresponding middle-sector derivatives are both approximately 1.2×10^{-7} . The critical-sector response is amplified by 1.24 and the trade-weighted aggregate response by 1.24. Holding the systemic term fixed gives an own-sector amplification of 1.24. Cross-sector spillovers account for 1.9 percent of the critical-sector amplification. The systemic channel is positive but quantitatively small in this calibration.

3.2 Folds and the sectoral cascade

Proposition 3 (Nondegenerate folds). *Let the remaining sectors lie on locally unique branches and write the resulting scalar residual as $R_m(F_m, \delta) = \tilde{T}_m(F_m, \delta) - F_m$. A nondegenerate fold at (F_m^*, δ^*) satisfies*

$$R_m = 0, \quad R_{mF} = 0, \quad R_{mFF} \neq 0, \quad R_{m\delta} \neq 0. \quad (12)$$

Near this point, two equilibria exist on one side of δ^* and none exist on the other. A hysteresis band requires two such folds together with the boundary signs that generate a low and a high branch. If $\eta_a = 0$ and sectors differ only in λ_m^0 , fold hazards satisfy $\delta_m^* \lambda_m^0 = K$ for a common constant K . An isolated sector jump changes aggregate fragmentation by $w_m \Delta F_m$.

The fold theorem is local. The reported global band is a numerical property of the calibrated composite map. The solver verifies two changes in the number of transversal roots and checks

tangency at their limiting locations. Premium dispersion determines whether the best response becomes steep enough to meet the nondegeneracy conditions. At log dispersion 0.5, the critical-sector band is [11.4, 13.1] percent. Raising dispersion to 0.7 removes the band in the reported parameter range.

A sector carrying fifteen percent of cross-bloc trade can jump while the aggregate moves by only its trade weight times the sectoral change. The spectral radius $\varrho(J)$ measures distance to instability in the full system. A scalar sector slope is adequate only when off-diagonal entries are negligible.

At a high-fragmentation equilibrium the hazard is higher by $\delta_m \theta \Delta F_m + \eta_a \Delta F$. Firms respond to disruption risk, and their response raises that risk when remaining trade deters escalation.

3.3 Dynamics and hysteresis

Relationships are sticky: a fraction ρ of firms reconsider their sourcing each year, so the sector state evolves as

$$F_{m,t+1} = (1 - \rho) F_{m,t} + \rho T_m(F_{m,t}, F_t, \delta_t), \quad (13)$$

a monotone map in both the state and the hazard. Nothing locks a firm in: a reconsidering firm returns to cross-bloc sourcing whenever the threshold condition says to.

Proposition 4 (Temporary shocks, permanent networks). *Let the hazard follow a spike path: $\delta_t = \delta^{sp} > \delta_{hi}^*$ for T^{sp} periods, then $\delta_t = \delta^{term} \in (\delta_{lo}^*, \delta_{hi}^*)$ forever. There is a finite duration $\bar{T}(\delta^{sp}, \rho)$, decreasing in the overshoot and in the reconsideration rate, such that the economy converges to the high-fragmentation equilibrium of δ^{term} if $T^{sp} \geq \bar{T}$, and returns to the low one if $T^{sp} < \bar{T}$. Reversal from the high branch requires the hazard to fall below δ_{lo}^* , while the band $[\delta_{lo}^*, \delta_{hi}^*]$ is a region of history dependence.*

The proposition is a result about the partial-adjustment rule in (13). Static multiplicity means that identical fundamentals admit several sourcing equilibria. Dynamic hysteresis here means that the predetermined state crosses the unstable fixed point. Expectational selection allows forward-looking firms to coordinate on a branch without that crossing. Physical irreversibility would arise from sunk investment or relationship capital and is absent from the baseline. Appendix C explains why these mechanisms need not select the same path.

I also solve the transition under perfect foresight. Let A_{mt} denote the present value of disruption losses from remaining exposed and let B_{mt} denote the present value of one unit of the switching premium. For an anticipated path, define

$$\begin{aligned} q_{mt} &= \beta_f (1 - \rho) (1 - \pi_{mt}), \\ A_{mt} &= \pi_{mt} \lambda_{mt} - \tau_t + q_{mt} A_{m,t+1}, \\ B_{mt} &= (1 - \chi F_{mt}) (1 - \varsigma_t) + q_{mt} B_{m,t+1}. \end{aligned} \quad (14)$$

The forward-looking threshold is $g_{mt}^{PF} = A_{mt}/B_{mt}$. The state path combines this threshold with the same reconsideration law as (13). In a stationary environment both present values divide

by $1 - q_m$, so their ratio equals the static threshold in (9). The numerical path solver iterates backward thresholds and forward shares until the maximum path residual falls below 9.8×10^{-12} .

The model is now complete as positive economics. Its normative content, who should pay whom at the margin, turns out to depend on where the sector sits relative to the fold, and that is the next section.

4 Welfare and Policy

4.1 The planner's margin and the sign of the wedge

Take a planner who can choose the sector's fragmented share, implementing it with threshold allocations, and who counts resource costs: premia actually paid, and disruption losses actually expected. Per unit of sector trade value,

$$B_m(F) = (1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m + (1 - F) \pi_m(F) \lambda_m(F), \quad (15)$$

where $q(F)$ is the premium of the marginal switcher. Differentiating and evaluating at the decentralized equilibrium, where the marginal switcher is privately indifferent, the private terms cancel and what remains is the net external effect of one more exit:

$$\tau_m^*(F) = \underbrace{(1 - F)[\lambda_m(F) \delta_m \theta + \pi_m \lambda_m^0 \xi]}_{\text{hazard and loss imposed on stayers}} - \underbrace{\chi (1 - f_m^0) \int_0^{q(F)} g dH_m}_{\text{premium relief delivered to switchers}}. \quad (16)$$

Proposition 5 (The wedge and the independence value). *$\tau_m^*(F) > 0$ at $F = f_m^0$ whenever $\theta > 0$. If $\chi > 0$, then $\tau_m^*(F) < 0$ as $F \rightarrow 1$, so a sign change exists. Let v_m be the government's flow value of one more unit of independent sourcing. The net wedge on exit is $\tau_m^*(F) - v_m$. A maintenance subsidy is corrective if and only if $v_m < \tau_m^*(F)$. Under the single-crossing condition verified for the reference scenario, the zero of τ_m^* is unique.*

An exit raises the disruption hazard on remaining links and thins the routes available to firms that stay. It also expands friendly capacity. The former forces make decentralized exit excessive. The capacity effect works in the opposite direction. In the reference scenario, τ_m^* is +6.4 percent at $\delta = 0.10$ and -4.9 percent at $\delta = 0.24$. These are conditional policy wedges before subtracting v_m .

4.2 Instruments

Policy can act on the switching premium, the disruption loss, or the return to maintaining trade.

A *friend-shoring subsidy* pays share ς of the switching premium. It shifts the threshold to $\hat{g}_m/(1 - \varsigma)$: mechanically it raises F_m , erodes the hostage, raises π_m , and drags both fold points toward the economy. It is the corrective instrument only where $\tau^* < 0$, past the flip. Deployed

Table 1: Calibration (annual).

Parameter		Value	Category	Basis
Trade weights	w_m	0.15 / 0.35 / 0.50	scenario types	critical / middle / flexible
Substitution elasticities	σ_m	2 / 4 / 8	external range	HS-level estimates (BW)
Cross-bloc input shares	s_m	0.50 / 0.35 / 0.20	scenario types	OECD ICIO moments
Disruption durations	D_m	1.5 / 1.0 / 0.5 yr	external range	published case studies
Implied losses	λ_m^0	150% / 15% / 1.6%	model implied	equation (1)
Median premia	$\bar{g}_m$	0.20 / 0.15 / 0.10	scenario	premium-design target
Premium dispersion	s	0.5	scenario	0.3 to 1.1 sensitivity
Pre-existing diversification	f_m^0	0.05 / 0.08 / 0.10	scenario	pre-period trade shares
Deterred share	θ	0.6 {0.4–0.7}	scenario range	sign from MMT
Thinning, capacity	ξ, χ	0.4, 0.4 {0–0.6}	scenario range	empirical targets
Systemic feedback	η_a	0.01	scenario	reported decomposition
Reconsideration rate	ρ	0.25	external proxy	four-year links (BS)
Low dispute hazard	δ^{pre}	4%	scenario	full axis reported
High dispute hazard	δ^{post}	10%	scenario	full axis reported
Cross-bloc intermediates / GDP	α	4%	public aggregate	trade and IO data

Notes: BW is Broda and Weinstein (2006). MMT is Martin et al. (2008). BS is Barrot and Sauvagnat (2016). Braces give the ranges for reference inputs.

below the fold it has the wrong sign, and deployed near the fold it can tip the sector outright, a possibility Section 5 prices at the calibration.

A *stockpile or buffer* cuts the loss a disruption inflicts: λ_m^0 falls by the covered fraction. It lowers thresholds, lowers F_m , preserves deterrence, and pushes the folds away. In the model’s accounting it is the main resilience instrument because it purchases exactly the object firms fear, disruption cost, while preserving the object society needs, the hostage. Its resource cost prices the comparison in practice.

A *maintenance subsidy* of τ^* on cross-bloc trade in critical inputs implements the planner’s optimum directly. Its political economy is hard, subsidizing trade with the adversary in the name of security, and the model supplies the logic: the subsidy buys deterrence, which is a public good priced at zero by every private contract.

The independence value v_m is not calibrated from observed subsidies. Proposition 5 instead reports the threshold. At the reference state, a zero maintenance subsidy requires $v_m \geq 6.4$ percent of sector trade value per year. This inequality is the policy result. The model does not assign a welfare value to independence.

5 Quantitative Analysis

5.1 Calibration

Table 1 separates external evidence from model-implied objects and reference inputs. The calculation maps these inputs into equilibrium objects. Section 6 supplies the design that estimates distance to a fold.

Equation (1) maps short-run substitution and duration into disruption losses. The critical-sector calculation of 150 percent represents zero emergency substitution over eighteen months.

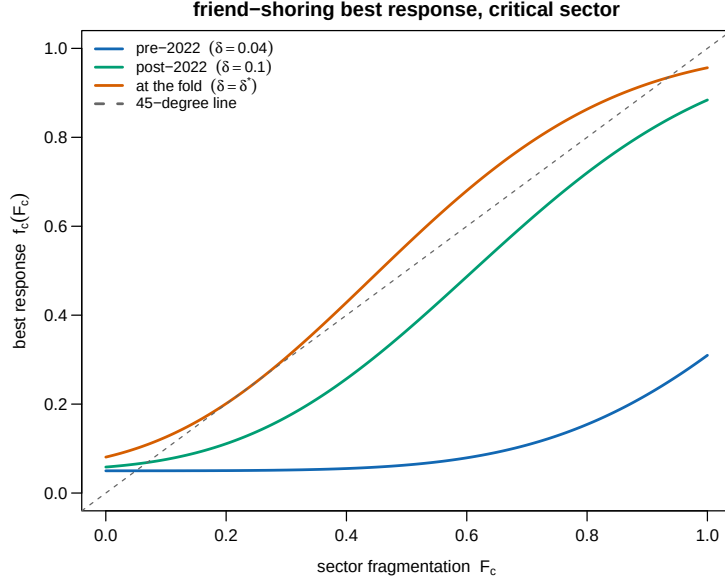


Figure 1: The critical sector’s best response at selected dispute hazards. At the fold, the curve is tangent to the diagonal and the low equilibrium disappears.

Equation (2) lowers this loss to 75 percent when inventory covers one quarter and emergency suppliers replace one quarter of the input. Half a year of inventory with one-half emergency substitution lowers it to 33.3 percent. The premium distribution and feedback elasticities then determine whether the composite best response develops a fold. The move from $\delta = 4$ percent to $\delta = 10$ percent defines the reference reassessment.

5.2 Reference multipliers and fold distance

The high-hazard reference equilibrium puts critical-sector fragmentation at 6.9 percent, against 5.0 percent in the low-hazard scenario. The aggregate moves from 8.6 to 8.8 percent. This comparative static isolates how heterogeneous criticality concentrates adjustment.

The calibrated composite map has a critical-sector band from 11.4 to 13.1 percent. Moving between the two reference hazards covers 66 percent of the distance from the low reference point to the upper fold. The sensitivity grid produces distances from 47.0 to 100.0 percent. Some cells have no fold and one cell lies beyond it at the high reference hazard. Four root grids with steps 0.004, 0.002, 0.001, and 0.0005 each recover three equilibria at the band midpoint. Their maximum root difference is 0. Newton refinement solves $R = R_F = 0$ at both folds with maximum equation residual 2.2×10^{-11} . The computed curvatures and transversality derivatives remain bounded away from zero.

Premium density at the switching margin governs the fold. Table 2 holds the median and interquartile range fixed while changing distribution shape. Every matched parametric family retains a fold, and the smoothed quantile sieve nearly reproduces the baseline band. Wider dispersion remains economically decisive. Raising log dispersion from 0.5 to 0.7 removes the fold

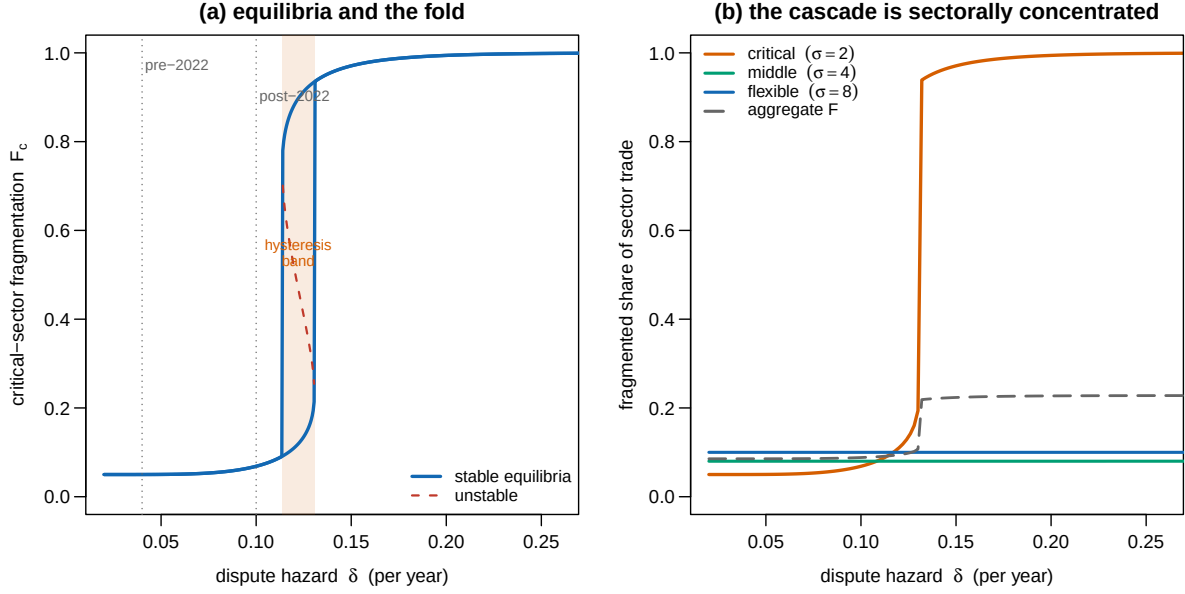


Figure 2: Panel (a): equilibria of the critical sector by dispute hazard. Solid branches are stable, dashed unstable, and the shaded region is the hysteresis band $[\delta_{lo}^*, \delta_{hi}^*] = [11.4\%, 13.1\%]$. Panel (b): fragmented shares by sector along the same axis, continuing from low initial conditions: the cascade is confined to the critical sector.

Table 2: Fold robustness across switching-premium distributions.

Premium cdf	Lower fold	Upper fold
Lognormal	11.37%	13.06%
Log-logistic	11.35%	13.01%
Weibull	10.76%	12.53%
Smoothed quantile sieve	11.36%	13.09%

Notes: each parametric cdf matches the baseline median and interquartile range. The sieve averages narrow kernels over 99 premium quantiles. Each row solves the full coupled-sector fixed point.

throughout the reported hazard range.

5.3 Hysteresis: a temporary confrontation, a permanent network

Figure 3 compares two economies that end at $\delta = 12.2$ percent. One reaches that value directly and converges to 12 percent fragmentation. The other spends twelve years at $\delta = 20.1$ percent and converges to 90 percent after the hazard returns. The spike moves the state across the unstable equilibrium. Perfect foresight preserves the two terminal selections at 12 and 90 percent. Anticipation changes the timing sharply on the spike path, with a maximum difference of 53.7 percentage points from adaptive adjustment. The high branch raises annual resource burden by 4.4 percent of critical-sector trade value, equal to 2.7 basis points of GDP under the trade-share calibration.

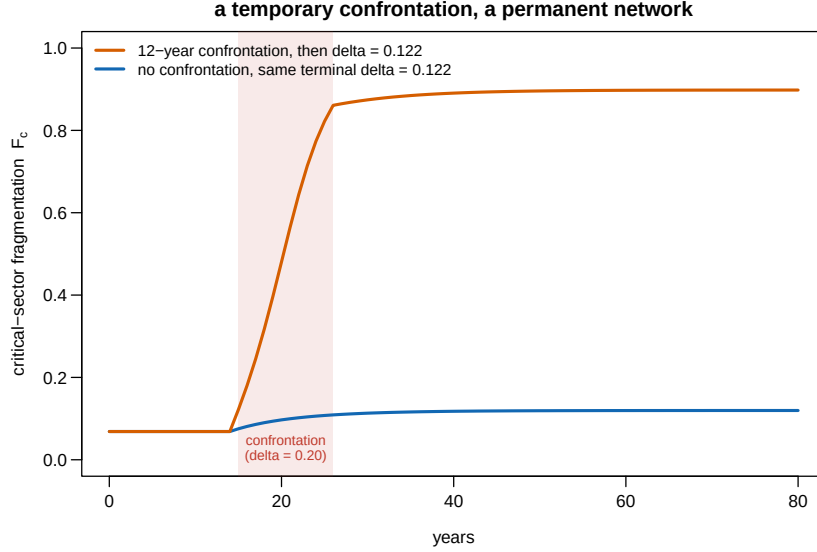


Figure 3: Two economies facing the identical terminal dispute hazard $\delta = 12.2\%$, inside the band. One lived through a twelve-year confrontation at $\delta = 20.1\%$. The other entered directly at the terminal hazard.

5.4 Policy at the calibration

In the reference scenario, a subsidy covering 25 percent of the switching premium moves the upper fold to 9.7 percent and selects a high-fragmentation equilibrium at the high hazard. A stockpile that reduces disruption loss by one third moves the upper fold to 19.7 percent. These rankings are conditional on v_m , the cost of the instruments, and the estimated deterrence relation. Figure 5 reports the model wedge before those additional terms are included.

6 Empirical Designs

The empirical unit is the publicly observed importer-exporter-HS6 relationship. Public data discipline the direct reallocation response and each component of the Jacobian separately. Joint estimation of those moments then determines distance to a fold.

Direct reallocation. Product-level bilateral trade from BACI (CEPII, 2026a) at HS6 enters the specification

$$\text{TradeShare}_{ijpt} = \alpha_{ijp} + \alpha_{ipt} + \alpha_{jpt} + \beta \text{Post}_t \text{GeoDist}_{ij} \text{Criticality}_{ip} + \varepsilon_{ijpt}, \quad (17)$$

where the research design constructs all exposure variables before 2022. Criticality combines short-run substitution evidence, OECD ICIO input shares, and public import-dependence measures. The coefficient describes the joint 2022 shift in war risk, sanctions, commodity conditions, and industrial policy. The 2018 tariff episode supplies a comparison with persistent supply investment.

Relationship exit and re-entry. Define $\text{Link}_{ijpt} = \mathbf{1}\{\text{Trade}_{ijpt} > \underline{v}\}$ using a threshold stated before estimation. I estimate exit and re-entry hazards at the importer-exporter-product level.

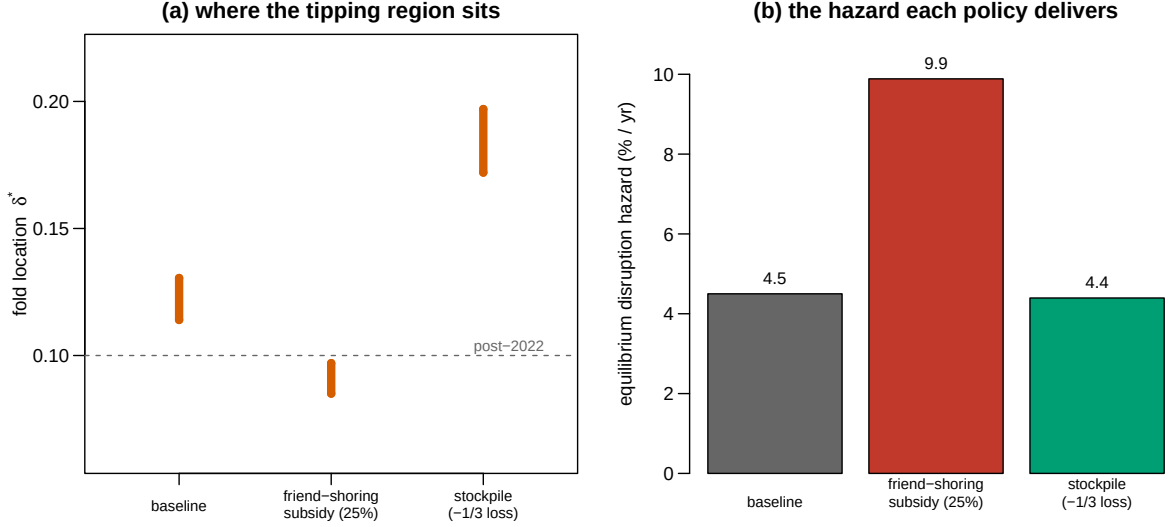


Figure 4: Policy scenarios. The dashed line is the high reference hazard. Panel (b) reports the equilibrium disruption hazard at $\delta = 10\%$.

Slow re-entry reflects partial adjustment, sunk relationship capital, persistent expectations, or basin dependence. Its interaction with independently estimated fold proximity identifies the model's distinct restriction.

Deterrence. A public dispute chronology completes the GSDB record with non-escalations. The deterrence equation conditions on that risk set at the dyad-year level:

$$\Pr(\text{Escalation}_{ij,t} = 1 \mid \text{Dispute}_{ij,t} = 1) = \Lambda(\alpha_{ij} + \tau_t + \beta \widehat{\text{TradeExposure}}_{ij,t-1} + \Gamma' X_{ij,t-1}), \quad (18)$$

where gravity variables and historical determinants predict trade exposure. Realized trade responds to political relations. Conditioning on observed disputes aligns the regression with the model's separation between dispute arrival and escalation. This design identifies the dyad-level trade response of escalation. The quantitative exercise maps that estimate into an identified set for sector deterrence. An official product-level export-control chronology supplies the risk set required for a sector point estimate of θ_m .

Premiums and capacity. Unit-value gaps supply noisy indicators of the latent switching premium. Quality, markups, freight, tariffs, supplier composition, and selection all enter the observed gap. I use several price indicators Z_{ijpt}^r in the measurement system

$$Z_{ijpt}^r = a_r + b_r g_{ijpt} + d_r' W_{ijpt} + u_{ijpt}^r, \quad (19)$$

where W_{ijpt} contains tariffs, freight proxies, quantities, and product-specific global prices. One loading fixes the scale. BACI unit values, CEPII TradePrices indices, and tariff-adjusted landed-cost gaps provide separate indicators. A sieve estimate of H_m propagates measurement

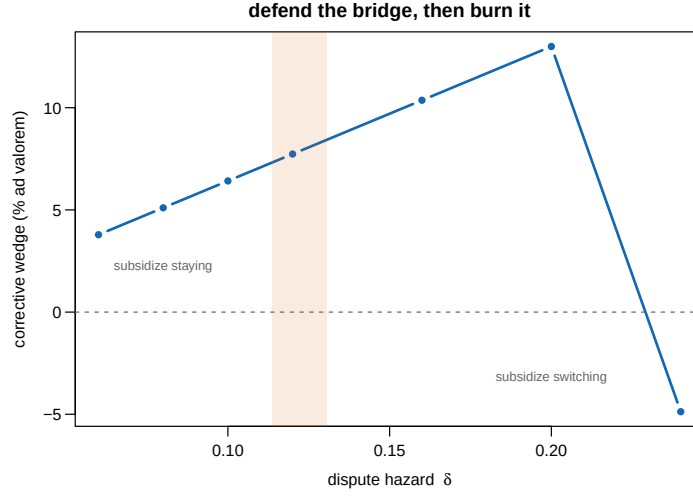


Figure 5: The corrective wedge τ^* by dispute hazard: positive (subsidize maintained cross-bloc trade) below and inside the band, negative (subsidize switching) beyond it.

uncertainty into the fold calculation. The capacity elasticity comes from the response of the latent premium to cumulative aligned demand. Third-country demand supplies shift-share variation, with product-specific global shocks entering directly as controls.

Route thinning. IMF PortWatch disruptions link recovery time to pre-event corridor thickness. Active exporter counts, shipment value, and alternative-route capacity measure thickness. This design identifies the maritime corridor mechanism. Export controls operate through licensing, legal substitution, and technology restrictions. I estimate their recovery technology in a separate event panel built from official control lists. A specification test determines whether the two estimates share a common ξ .

The fixed-point solver receives the joint empirical bootstrap. Each draw records $\varrho(J)$ and classifies the equilibrium as no fold, a fold above the feasible hazard range, or a fold within that range. The bootstrap frequency of each class supplies the reported probability. Conditional on a fold, the same draws report both locations and the high-branch welfare ranking. Alternative parametric cdfs and the nonparametric sieve enter as separate model-selection rows. One-at-a-time sensitivity remains a diagnostic.

The deterrence regression disciplines the dyad-level escalation externality. Relationship persistence joins the price measurement system, capacity response, and route evidence in the structural fit. The sector deterrence identified set carries directly into the fold probabilities.

Every dataset in the empirical program is public. Table 4 records the implementation.

7 Conclusion

Decentralized sourcing places the trade-conflict relation inside firms' private decisions. An exit reduces the surplus that escalation would destroy and raises the disruption hazard on remaining

Table 3: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
P1: cross-bloc shares fall with exposure	conditional sign	0	BACI triple difference
P2: response dose-monotone in constructed criticality λ_m^0	ordering of (1)	flat	criticality interactions
P3: re-entry slows near independently measured criticality	interaction sign	no interaction	relationship hazards
P4: escalation hazard falls with predicted trade exposure	$\beta < 0$	$\beta = 0$	dyad-dispute panel
P5: aligned-supplier premia fall with cumulative demand	$\chi > 0$	$\chi = 0$	unit-value gaps
P6: local persistence rises with the leading eigenvalue	$\varrho(J) \uparrow$	constant	sector dynamics

Notes: the table reports sign and interaction restrictions. Scenario magnitudes are not used as empirical targets unless separately identified.

links. In a multisector economy, the Jacobian of best responses governs equilibrium amplification. Sectoral criticality alone cannot determine the multiplier.

The calibrated economy satisfies nondegenerate tangency and the global crossing conditions. Root enumeration, grid refinement, and fold residuals establish the numerical branch structure. Perfect foresight accelerates the transition but preserves terminal branch selection in the confrontation experiment. Inventory and emergency sourcing move the fold away by reducing the short-run loss that motivates exit.

Policy depends on the government's value of independent sourcing. A maintenance subsidy is corrective when $v_m < \tau_m^*$. Deterrence and route thickness raise τ_m^* . Friendly capacity lowers it. The empirical design measures these forces separately and carries their joint uncertainty into the probability of a feasible fold and the welfare ranking of its high branch.

Table 4: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or prox- ied	Measure
CEPII BACI	HS6 bilateral trade	annual	public	observed	in hand
CEPII GeoDep	dependency and concentration	annual	public	observed	in hand
CEPII MAcMap	applied tariffs at HS6	periodic	public	observed	in hand
OECD ICIO	cost shares and cen- trality	annual	public	observed	in hand
HS-level elasticities	Broda and Wein- stein (2006)	fixed	published	estimated, pre- period	in hand
IMF PortWatch	maritime disrup- tions	event	public	observed	in hand
GSDB release 4	through 2023	episode	public request	observed	obtainable
MID or UCDP events	dyad disputes	annual	public	observed	in hand
UN ideal points	all members	annual	public	observed	in hand
CEPII TradePrices	product prices	annual	public	price proxy	in hand
BIS Federal Register no- tices	export-control rules	event	public CSV	observed	in hand

Notes: the price measurement system treats BACI unit values as one indicator because quality and composition vary. TradePrices and landed-cost gaps add distinct indicators. Dated BIS Federal Register notices supply the product-control chronology (U.S. Bureau of Industry and Security, 2026).

A Proofs

A.1 Proof of Lemma 1

Let $V^{sw} = -g^{\text{eff}}/(1 - \beta_f)$ be the value of the absorbing friend-shored state and V^{st} the value of staying, which pays the expected loss and transits to the absorbing state upon disruption: $V^{st} = -\pi\lambda + \beta_f[(1 - \pi)V^{st} + \pi V^{sw}]$. Solving, $V^{st}(1 - \beta_f(1 - \pi)) = -\pi\lambda + \beta_f\pi V^{sw}$, and $V^{st} \geq V^{sw}$ iff $-\pi\lambda + \beta_f\pi V^{sw} \geq (1 - \beta_f(1 - \pi))V^{sw}$ iff $-\pi\lambda \geq (1 - \beta_f)V^{sw} = -g^{\text{eff}}$, which is the flow comparison. No discount-rate term survives. ■

A.2 Deterrence microfoundation

Lemma 2 (Linear hazard from escalation choice). *Suppose disputes arrive with probability δ_m and the adversary escalates when $b \geq \mathcal{C}_m$. Then the dispute component of the link hazard is (4) without $\eta_a F$. It is increasing in each fragmentation state whenever $g_m^b(\mathcal{C}_m) > 0$ and remaining trade raises $\mathcal{C}_m$. If b is uniform and the escalation probability is interior, the hazard is affine.*

The escalation probability is $1 - G_m(\mathcal{C}_m)$. Differentiation gives $-g_m^b(\mathcal{C}_m)\mathcal{C}_m F_n > 0$. With a uniform distribution on $[0, \bar{b}]$, the interior probability is $1 - \mathcal{C}_m/\bar{b}$. Multiplication by δ_m and normalization of its intercept give the dispute component of (5). The separate term $\eta_a F$ preserves monotonicity. ■

A.3 Proof of Proposition 1

Under the affine hazard, write

$$\hat{g}_m(F_m, F) = \lambda_m^0 \frac{\delta_m[(1 - \theta) + \theta F_m] + \eta_a F}{1 - \chi F_m} (1 + \xi F_m).$$

Holding F fixed, the product and quotient rules give

$$\frac{\partial \hat{g}_m}{\partial F_m} = \lambda_m^0 \left[\frac{\delta_m \theta (1 + \xi F_m)}{1 - \chi F_m} + \frac{\xi \pi_m}{1 - \chi F_m} + \frac{\chi \pi_m (1 + \xi F_m)}{(1 - \chi F_m)^2} \right].$$

This is the own-sector formula in the proposition. Holding F_m fixed gives

$$\frac{\partial \hat{g}_m}{\partial F} = \frac{\eta_a \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m} > 0$$

when $\eta_a > 0$. Under the general hazard, the same sign follows from (6) and the positive density of H_m . If all feedback parameters vanish, T_m is independent of $\mathbf{F}$ and the fixed point is unique. ■

A.4 Proof of Proposition 2

Continuity of $\mathbf{T}$ on the compact convex product set gives existence by Brouwer's theorem. Differentiating $\mathbf{F}^* = \mathbf{T}(\mathbf{F}^*, z)$ gives $(I - J)d\mathbf{F}^*/dz = \mathbf{b}_z$. If $\varrho(J) < 1$, the inverse exists and

equals $\sum_{k=0}^{\infty} J^k$. Proposition 1 makes J nonnegative, so every term of the inverse is nonnegative. Setting off-diagonal entries to zero gives the scalar special case. ■

A.5 Proof of Corollary 1

Within a sector and equilibrium, variation in an idiosyncratic shifter moves individual thresholds while holding $\mathbf{F}$ fixed. Such variation identifies elements of $\mathbf{b}_z$. Variation across sectors also changes H_m , λ_m^0 , and aggregate loadings, so the network component need not cancel. Recovering J requires state variation or structural restrictions on these loadings. ■

A.6 Proof of Proposition 3

At a point satisfying (12), the second-order expansion is

$$0 = R_{m\delta}(\delta - \delta^*) + \frac{1}{2}R_{mFF}(F_m - F_m^*)^2 + o(|\delta - \delta^*| + |F_m - F_m^*|^2).$$

The nonzero coefficients imply two nearby roots on one side and none on the other. A global band follows only when two folds are present and the residual has the stated boundary signs. With $\eta_a = 0$ and common non-loss primitives, the threshold depends on δ and λ_m^0 through their product. Hence $\delta_m^* \lambda_m^0$ is constant across sectors. The aggregate jump follows from $F = \sum_m w_m F_m$. ■

A.7 Proof of Proposition 4

The map $F \mapsto (1 - \rho)F + \rho T(F, \delta)$ is increasing in F and δ , so paths from ordered initial conditions stay ordered, and with constant δ the path is monotone toward the nearest fixed point in its direction of motion. During the spike, $\delta^{sp} > \delta_{hi}^*$ implies a unique fixed point $F^{hi}(\delta^{sp})$ exceeding the terminal unstable point $U(\delta^{term})$. The path rises toward it and crosses $U(\delta^{term})$ at some finite $\bar{T}$, decreasing in δ^{sp} (which raises the map) and in ρ (which speeds it). If reversion occurs after $\bar{T}$, the state exceeds $U(\delta^{term})$, where $T(F, \delta^{term}) > F$ holds on (U, F^{hi}) : the path rises monotonically to the high branch. If before, the state is below U , where $T < F$: it falls back. From the high branch, δ must fall below δ_{lo}^* , where the high fixed point is destroyed, before any monotone path can leave its basin. ■

A.8 Proof of Proposition 5

Differentiate (15) in F , using $(1 - f_m^0)h_m(q)q'(F) = 1$. The premium term gives

$$\frac{d}{dF} \left[(1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m \right] = (1 - \chi F)q(F) - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m.$$

The exposure term gives

$$\frac{d}{dF} [(1 - F)\pi_m(F)\lambda_m(F)] = -\pi\lambda + (1 - F) [\lambda \delta_m \theta + \pi \lambda_m^0 \xi].$$

At the decentralized equilibrium the marginal switcher is indifferent, $(1 - \chi F)q(F) = \pi\lambda$, so the private terms cancel. Therefore

$$\frac{dB_m}{dF} = (1 - F) \left[\lambda \delta_m \theta + \pi \lambda_m^0 \xi \right] - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m = \tau_m^*(F).$$

A positive τ_m^* means one more exit raises social burden, so decentralized exit is excessive and the corrective instrument subsidizes maintained trade. At $F = f_m^0$ the integral vanishes and $\tau^* = (1 - f_m^0)[\lambda \delta \theta + \pi \lambda^0 \xi] > 0$ when $\theta > 0$. As $F \rightarrow 1$, the positive bracket vanishes while the capacity term remains negative when $\chi > 0$. Continuity gives a crossing. A government value $v_m F$ subtracts v_m from the marginal burden, giving the net wedge $\tau_m^* - v_m$. A payment on maintained trade shifts the private comparison by the same amount and implements the threshold allocation. ■

B Microfoundations

The CES loss. Let the firm's input bundle be CES over the cross-bloc variety and a composite substitute. Removing a variety with initial cost share s_m multiplies the price index by $(1 - s_m)^{1/(1-\sigma_m)}$. With $\sigma_m = 2$ and $s_m = 1/2$, the index doubles. This endpoint calculation applies to the uncovered share during the uncovered duration. Inventory I_m and emergency substitution e_m deliver equation (2). Downstream input-output propagation scales the direct loss through sector centrality (Baqaee and Farhi, 2019; Carvalho et al., 2021). The calibration prices the direct loss and the network propagation separately.

The escalation game. Lemma 2 treats one dispute. The same argument applies to repeated disputes so long as the escalation choice is static ex post, which it is here because portfolios are predetermined within the period. A forward-looking adversary, for whom today's escalation accelerates tomorrow's fragmentation and erodes future hostage-taking capacity, adds a second time-inconsistency problem on the government side. The present specification keeps the demand side clean and puts all strategic complementarity in firms' sourcing choices.

Capacity scale. The compression factor $(1 - \chi F_m)$ stands for entry and learning in the friendly supply base. With an in-bloc supply curve of elasticity $1/\chi$ facing cumulative demand proportional to F_m , the marginal premium falls linearly to first order. The empirical design therefore loads price compression on cumulative reallocation rather than contemporaneous volume.

C History versus Expectations

The adaptive rule (13) lets reconsidering firms use the current state as their forecast. A forward-looking firm instead compares present values along an anticipated path. Suppose its current relationship survives to the next date when the firm neither reconsiders nor suffers disruption.

This event has discounted probability $q_{mt} = \beta_f(1 - \rho)(1 - \pi_{mt})$. The expected loss component and premium component obey (14). Backward recursion yields $g_{mt}^{PF} = A_{mt}/B_{mt}$. Forward iteration then gives

$$F_{m,t+1} = (1 - \rho)F_{mt} + \rho \left[f_m^0 + (1 - f_m^0)H_m(g_{m,t+1}^{PF}) \right]. \quad (\text{C.1})$$

The algorithm alternates the backward recursion and (C.1). Damping produces a path residual below 9.8×10^{-12} in every reported experiment. At a stationary path, $A_m = (\pi_m \lambda_m - \tau)/(1 - q_m)$ and $B_m = (1 - \chi F_m)(1 - \varsigma)/(1 - q_m)$. Their ratio equals the static threshold. Lemma 1 therefore holds exactly at every stationary equilibrium.

Forward-looking behavior changes transition speed because expected capacity growth lowers future switching premia. In the direct terminal-hazard path, the maximum critical-sector difference from adaptive dynamics is 1.3 percentage points. In the confrontation path it reaches 53.7 percentage points. Both methods converge to the same low and high terminal branches, respectively. Thus anticipation matters for timing in this calibration while the unstable equilibrium continues to determine long-run selection after the temporary spike.

D Numerical Continuation and Root Audit

For each hazard, the algorithm fixes a candidate critical-sector share and solves the remaining sector equations by damped contraction. This step produces the aggregate state as a function of the candidate share. Substitution into the critical-sector equation gives the scalar residual $R_1(F_1, \delta)$. The root routine evaluates this residual at both boundaries and on the entire unit interval. It brackets every sign change and refines each bracket by bisection. The continuation routine repeats this enumeration along the hazard grid and classifies stability from the derivative of the composite map.

The fold solver starts from the last three-root bracket and jointly solves $R_1 = 0$ and $R_{1F} = 0$. Centered differences provide R_{1F} , $R_{1\delta}$, and R_{1FF} . A damped Newton step preserves the unit interval and lowers the maximum equation residual at every iteration. The reported lower and upper folds have maximum residual 2.2×10^{-11} . Their curvature and transversality derivatives exceed 2.1 and 3.9 in absolute value. Root enumeration at the band midpoint returns three roots on grid steps 0.004, 0.002, 0.001, and 0.0005, with maximum location difference 0.

The uncertainty routine applies this complete root audit to every bootstrap draw. It records a no-fold outcome when the residual has one transversal root throughout the feasible hazard interval. A fold above the feasible interval forms a separate outcome. The remaining draws supply the joint distribution of fold locations and branch welfare. This classification gives no-fold probability a direct sampling interpretation.

E Parameter Discipline and Sensitivity

Table 5 records parameter status and ranges. The no-feedback benchmark sets $\theta = \xi = \chi = \eta_a = 0$. Fold existence varies jointly with the feedback block and premium dispersion. The empirical

bootstrap converts the estimated joint parameter distribution into fold-class probabilities.

Table 5: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source	Range examined	Effect of the range
$\sigma_m, s_m, D_m, I_m, e_m$	eqs. (1), (2)	elasticity, share, years	short-run disruption studies, ICIO	buffered cases	fix λ_m^0 , hence tipping order
λ_m^0	eq. (9)	ad val. flow	implied by eq. (2)	1.6% to 150%	fold position scales as $1/\lambda_m^0$
$\bar{g}_m, s$	eq. (10)	ad val. orem, log pts	scenario, premium log design	s 0.3 to 1.1	fold absent above $s \approx 0.7$
f_m^0	eq. (10)	share	scenario, pre-period target	floor check	level floor only
θ	eq. (5)	share deterred	scenario, deterrence design	0.4 to 0.7	fold location and Jacobian
ξ, χ	eqs. (7), (8)	elasticities	scenario, separate designs	0 to 0.6	fold existence and policy wedge
η_a	eq. (5)	per year	scenario	0 to 0.03	off-diagonal spillovers
ρ	eq. (13)	per year	Barrot and Sauvagnat (2016)	timing check	timing of $\bar{T}$ only
$\delta^{pre}, \delta^{post}$	eq. (5)	per year	scenario	full axis in figures	positions reference states
v_m	Prop. 5	ad valorem flow	policy input	threshold reported	maintenance-subsidy sign
w_m, α	aggregation	shares	trade aggregates	scale check	scale GDP numbers

Notes: the empirical implementation uses short-run substitution estimates. Long-run trade elasticities enter the sensitivity analysis. The joint bootstrap propagates uncertainty after estimating the component moments.

F Data Appendix

Trade. CEPII BACI, HS-6 bilateral flows and unit values, 1995–2024 (CEPII, 2026a), with extensive-margin relationship panels constructed at exporter-importer-product level. Bloc and alignment measures from UN ideal points (Bailey et al., 2017), fixed at pre-2022 values.

Criticality. The baseline empirical construction uses short-run substitution estimates, OECD ICIO cost shares, CEPII GeoDep concentration, and event-level recovery durations from Port-Watch (OECD, 2026; CEPII, 2024; International Monetary Fund, 2023). The long-run elasticities in Broda and Weinstein (2006) provide a sensitivity bound rather than the preferred short-run input. Every component is dated before the event used for validation.

Disputes and escalations. GSDB release 4 supplies sanctions episodes through 2023 (Global Sanctions Data Base, 2025). A public MID or UCDP-type chronology supplies the dispute risk set. The research team codes escalation at the dyad-year level. An assembled and validated official-list chronology supports product-level deterrence estimates.

Prices and tariffs. BACI unit values are compared with CEPII TradePrices. MAcMap-HS6 supplies bilateral applied tariffs at the product level (CEPII, 2026c; CEPII, 2026b).

Fragmentation. The BACI extensive margin measures $F_{m,t}$ as the share of sector m 's pre-2022 cross-bloc trade value in sustained relationship exits. This is the empirical counterpart of the model's state variable.

G Extensions

Connector countries. Let measured friend-shoring reroute content through politically intermediate countries. Connectors lower the observed premium and raise fragmentation of direct links. If rerouted trade still creates surplus that escalation would destroy, measured bilateral fragmentation overstates deterrence erosion. Direct-trade and value-added exposure measures distinguish these cases (Gopinath et al., 2025; Freund et al., 2023).

Two-sided fragmentation. Making both blocs' firms choose adds the adversary's exits to the hostage erosion. With symmetric parameters the sector map's slope doubles where deterrence is concerned, moving folds closer. The one-sided model is thus conservative about tipping.

Correlated disruption and percolation. Here a severed link harms only its firm. In production networks link shocks propagate (Elliott et al., 2022; Baqaee and Farhi, 2019), making λ depend on the pattern as well as the level of fragmentation. The interaction of strategic hazard with percolation fragility is unexplored and, I suspect, reinforcing.

The vulnerability motive. If the government values rewired trade at v per unit beyond firms' losses, the wedge becomes $\tau^* - v$ and the planner's problem otherwise stands. All sign results survive for $v < \tau^*$, and the policy discussion in Section 4 prices the threshold.

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Table 6: Sensitivity: the critical sector’s fold band $[\delta_{lo}^*, \delta_{hi}^*]$ by deterrence share θ and capacity elasticity χ , in dispute-hazard units. “Unique” means a unique equilibrium at all hazards.

	$\chi = 0.0$	$\chi = 0.2$	$\chi = 0.4$	$\chi = 0.6$
$\theta = 0.4$	none	none	0.107–0.107	0.086–0.099
$\theta = 0.5$	none	none	0.113–0.117	0.087–0.110
$\theta = 0.6$	none	0.138–0.141	0.116–0.133	0.088–0.127
$\theta = 0.7$	0.165–0.171	0.144–0.163	0.119–0.157	0.090–0.151

Figure 6 evaluates 165 scenario cells. The existence boundary slopes downward because deterrence and capacity feedback substitute in the best-response slope. At $\theta = 0.6$, a fold appears when $\chi \geq 0.15$. The grid contains 48 cells with a unique branch, 117 with a fold above the high reference hazard, and 16 with a fold below it. These counts summarize the scenario grid and carry no probability interpretation.

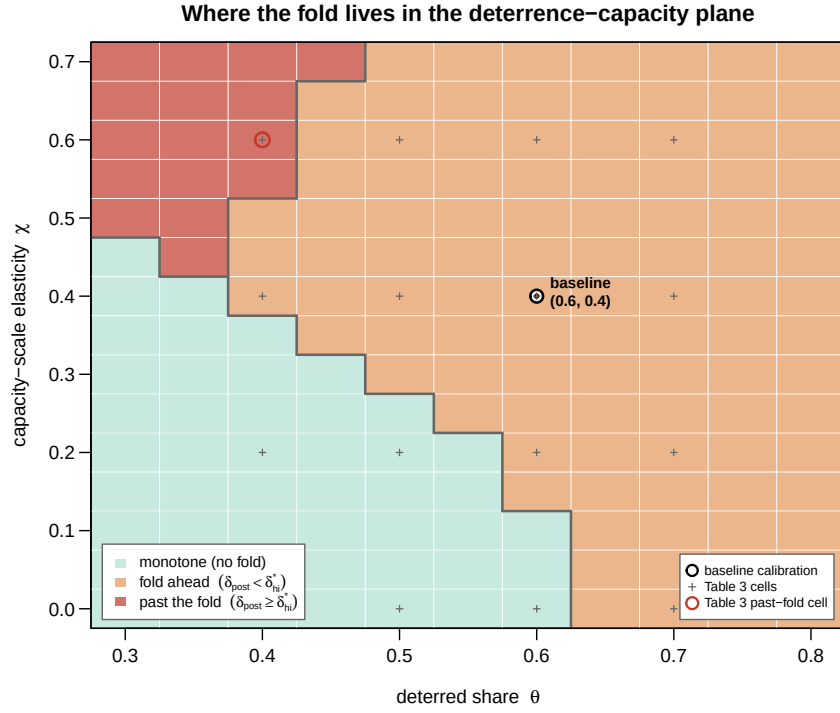


Figure 6: The fold in the (θ, χ) scenario plane. Shading classifies uniqueness, a fold above the high reference hazard, or a fold below it. Crosses mark the coarse-grid cells and the dot marks the reference scenario.